

Problem-Solution Fit

Date	29/09/26
Team ID	SWTID-2026-8504
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Source-based conversion of the EduGenie documentation into the supplied canvas.

1. CUSTOMER SEGMENT(S) [CS] - School / college students - Beginners & self-learners - Educational institutions - Content platforms	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Minimal hardware / low-footprint goal - Limited prior knowledge or technical experience - Need simple, accessible learning support - Works well on devices such as Mac M1	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing educational resources / static content. Pros: information is available. Cons: content may feel overwhelming and may not combine instant Q&A, quizzes, summaries and personalized paths.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Complex concepts can be difficult to understand. - Need quick academic answers. - Long passages need concise revision. - Need self-assessment and structured learning. Frequency: recurring study needs; not quantified in source.	9. PROBLEM ROOT / CAUSE [RC] - Traditional content may be overwhelming or inaccessible. - Learners have different levels of prior knowledge. - Support is needed across Q&A, explanation, revision, assessment and planning. - One-size-fits-all learning can limit personalization.	7. BEHAVIOR + ITS INTENSITY [BE] - Ask questions when doubts arise. - Request simpler explanations. - Generate quizzes to check understanding. - Summarize content for revision. - Request beginner-to-advanced learning paths.
3. TRIGGERS TO ACT [TR] - Difficult or unfamiliar topic - Need for a quick answer - Revision / self-assessment - Long passage to summarize - Starting a new topic / learning path	10. YOUR SOLUTION [SL] EduGenie — lightweight AI learning assistant. - Gemini 1.5 Pro: Q&A, summaries, quizzes, learning paths. - LaMini-Flan-T5-783M: simple explanations. - FastAPI backend + HTML/CSS frontend.	8. CHANNELS OF BEHAVIOR [CH] Intensity: recurring/task-driven; not quantified. ONLINE - EduGenie web interface - FastAPI application
4. EMOTIONS BEFORE / AFTER [EM] BEFORE - Content feels complex / overwhelming - Difficulty with unfamiliar concepts AFTER - Clearer understanding - Concise answers / summaries - Self-assessment and structured guidance		OFFLINE - Local Python/FastAPI environment - Local lightweight explanation model

Note: Frequency/intensity are marked as not specified where the source does not quantify them.